

CIA 1 Questions

Question 1: Smart Cold Chain Monitoring System using Cloud

Design and evaluate a Cloud-based Smart Cold Chain Monitoring System for monitoring the transportation and storage of temperature-sensitive products such as vaccines, pharmaceuticals, or perishable food items. The system should continuously collect data from IoT sensors (temperature, humidity, GPS location, and door status) and provide real-time monitoring, alerts, and analytics through a cloud platform.

Question 2: Smart Healthcare Remote Patient Monitoring Platform

Develop and critically evaluate a cloud-based Remote Patient Monitoring System for continuously monitoring patient vital signs such as heart rate, blood pressure, oxygen saturation, and body temperature. The system should provide alert to the patient through wearable device.

Question 3: Intelligent Smart City Traffic Monitoring using Cloud Computing

Design an intelligent Traffic Monitoring System that collects data from cameras and traffic sensors to monitor congestion and optimize traffic flow. The alert should be enabled to traffic controlling authorities.

Question 4: Industrial IoT Predictive Maintenance Platform

Design a cloud-based Predictive Maintenance System for monitoring industrial machines using vibration, temperature, and pressure sensors. The owner of the industry must receive alerts if there is a failure of sensors.

Question 5: Cloud-based Environmental Monitoring and Disaster Alert System

Design a cloud-enabled Environmental Monitoring System that continuously monitors rainfall, river water levels, air quality, and weather conditions to generate disaster alerts. The alert should reach the email and mobile numbers of registered users who are responsible to act.

Expected Deliverables (Common to all Questions)

1. System Architecture Diagram for your allocated question (5 Marks)
Cloud Architecture (AWS/Azure/GCP) – Use one of them
2. Creating Docker Image with explanation of steps you followed (5 Marks)
3. Kubernetes Deployment Architecture (2 Marks)
4. CI/CD Pipeline Workflow (2 Marks)
5. Evaluation Report with Design Justification (3 Marks)
6. Conclusion and Future Enhancements (3 Marks)

Note: Your report must include 1 to 6 items shown above in the same order and it should include screenshots wherever applicable.

You must containerize their IoT application using **Docker** and publish the **Docker image** to **Docker Hub**. This will demonstrate your ability to package and distribute applications using industry-standard containerization practices. The image should be publicly accessible, properly tagged, and capable of being deployed using Docker and Kubernetes without modification.